

Multiplexing

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Managing Redo log files

- All the changes made by the users are logged
- Fundamental for the recovery process
- At least 2 different groups of redo logs with at least one member in each

Multiplexing of Redologs

- Maintaining more than one member in each of your group
- All members of redo logs groups are identical
- Multiplexing is designed to protect against the loss of single copy of logfile
- LGWR simultaneously writes to all members of a group
- Multiplexing done in mount mode (recommended)
- Multiplexed redologs should ideally be located on different disk drives under different disk controller
- Oracle will continue to write to the surviving members of the online redo log groups when one member is not writable (perhaps due to the problem involving in different disk group)

Lab practice

- Adding and dropping the group
- Adding and dropping the members within a group
- Renaming the files or moving the location
- Redo log status:
 - UNUSED
 - CURRENT
 - ACTIVE
 - INACTIVE

Add and drop logfile

- alter database add logfile member
'C:\app\Administrator\product\18.0.0\oradata\logfiles\redo03_a.log'
to group 3;
- alter database drop logfile member
'C:\APP\ADMINISTRATOR\PRODUCT\18.0.0\ORADATA\LOGFILES\RED
O03_A.log';
- You cannot drop online redo log files if the log file is part of the
current group and active (if there is no extra surviving member)
 - alter database switch logfile;
 - Alter system checkpoint=> to make inactive (for checkpointing)

Add and drop redo group

- alter database add logfile group 4
'C:\APP\ADMINISTRATOR\PRODUCT\18.0.0\ORADATA\XE\REDO04.LOG' size 100M;
- alter database drop logfile group 4;
- You cannot drop active/current redo log group.
- At least 2 group should be there

Renaming or moving logfiles

- Alter database rename file 'source location' to 'destination'
- Can be done in mount mode only
- Cannot rename in open mode
- File move should be done in OS level in mount mode or shutdown mode

Control files and its multiplexing

Introduction

- Control files are key files that hold datafile names and location, and lot of other important information about database
- Database needs only one control file, but because this is such an important file, you can always save multiple copies of it to protect from single loss of file.
- Creating copies of control file to different location is called multiplexing of controlfiles
- Oracle recommend at least 2 control files per instance;
- If one control file is damaged or accidentally deleted, the instance will fail when it tries to access the control files.
- If you have multiple copies, you can always restart the database after first copying the undamaged control file to different location
- View: v\$controlfile;
- Contents of control files:
 - Database name & identifier, timestamp of database creation , tablespace and datafile names and location, redologs file name and location, current redo sequence number, checkpoint information, redolog archive info, backup info etc

STEPS of multiplexing the control files

- ALTER SPFILE WITH FOLLOWING COMMAND:
 - SQL> alter system set control_files =
'C:\APP\ADMINISTRATOR\PRODUCT\18.0.0\ORADATA\XE\CONTROL01.CTL','C:\APP\ADMINISTRATOR\PRODUCT\18.0.0\ORADATA\XE\CONTROL02.CTL','C:\app\Administrator\product\18.0.0\oradata\controlfile\CONTROL03.CTL' scope=spfile;
- Shutdown the database instance;
 - SQL> shutdown immediate;
- Use Operating system command to copy existing control file to location you specified for new file
 - CP 'source location' to 'destination location'
- Open the database
 - Startup;